

Hooke's law and simple harmonic motion

Basic skills

Q1. $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 18y = \cos 2x$

a) auxiliary for $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 18y = 0$

$$m^2 + 3m - 18m = 0$$

$$\Rightarrow (m+6)(m-3) = 0 \Rightarrow m = -6, 3$$

so $y = Ae^{-6x} + Be^{3x} \leftarrow CF$

need both
cos, sin
↓

b) AS trig try $y = a \cos 2x + b \sin 2x$

$$\frac{dy}{dx} = -2a \sin 2x + 2b \cos 2x$$

$$\frac{d^2y}{dx^2} = -4a \cos 2x - 4b \sin 2x$$

In ODE $\Rightarrow -22a \cos 2x - 22b \sin 2x - 6a \sin 2x + 6b \cos 2x = \cos 2x$

$$\Rightarrow 6b - 22a = 1, -6a - 22b = 0$$

$$\Rightarrow a = -\frac{22}{6}b \Rightarrow 6b + \frac{484}{6}b = 1$$

$$\Rightarrow \frac{550}{6}b = 1 \Rightarrow b = \frac{6}{550} \Rightarrow a = -\frac{22}{550}$$

$$\text{SO } y_{PI} = -\frac{22}{550} \cos 2x + \frac{6}{550} \sin 2x$$

c) Add CF, PI

$$y = Ae^{-6x} + Be^{3x} + \frac{1}{225} (3 \sin 2x - 11 \cos 2x)$$

$$d) y(0) = 0 \Rightarrow A + B - \frac{11}{225} = 0$$

$$\frac{dy}{dx} = -6Ae^{-6x} + 3Be^{3x} + \frac{1}{225} (6 \cos 2x + 22 \sin 2x)$$

$$\frac{dy}{dx}(0) = 0 \Rightarrow -6A + 3B + \frac{6}{225} = 0$$

$$\text{SO } \begin{aligned} 3A + 3B &= \frac{33}{225} \\ -6A + 3B &= -\frac{6}{225} \end{aligned}$$

$$\hline 9A = \frac{39}{225}$$

$$A = \frac{13}{675}$$

$$\text{SO } A = \frac{13}{675}$$

$$\begin{aligned} B &= \frac{11}{225} - A \\ &= \frac{4}{135} \end{aligned}$$

so Particular solution

$$y = \frac{13}{675} e^{-6x} + \frac{4}{135} e^{3x} + \frac{1}{225} (3 \sin 2x - 11 \cos 2x)$$

Q2.

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

→ as $v = \frac{dx}{dt}$

Q3. $4 \cos \theta + 5 \sin \theta = R \sin(\theta + \alpha)$

$$\sin(A+B) = \sin A \cos B + \sin B \cos A$$

$$\text{so } R \sin \alpha = 4, R \cos \alpha = 5$$

$$R^2 = 4^2 + 5^2 = 41 \Rightarrow R = \sqrt{41}$$

$$\tan \alpha = \frac{4}{5} \Rightarrow \alpha = \tan^{-1}\left(\frac{4}{5}\right) = 0.675$$

↑
Rads

$$\text{so } \sqrt{41} \sin(\theta + 0.675)$$

↑
Amplitude

↑
Phase

Magnitude $\sqrt{41}$, Phase shift 0.675

Q 4. $\ddot{x} = -\omega^2 x$

$\ddot{x} = \frac{d^2x}{dt^2}$

so $\ddot{x} + \omega^2 x = 0$ auxiliary $m^2 + \omega^2 = 0$

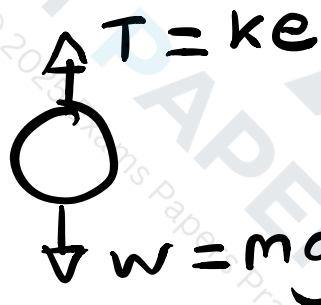
$\Rightarrow m = \pm \omega i$ so

$x(t) = A \cos(\omega t) + B \sin(\omega t)$

Competency skills

Q 5.

a)



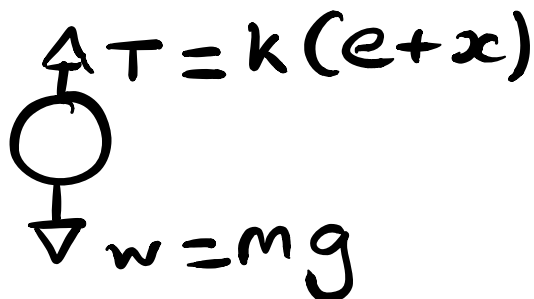
so resultant

$F = mg - ke$

As equilibrium $F = 0 \Rightarrow mg - ke = 0$

$\Rightarrow ke = mg$

b)



$F = mg - k(e+x)$

now $F = ma$ so $a = \frac{1}{m} F = g - \frac{k}{m} (e+x)$

as $a = \ddot{x} \Rightarrow \ddot{x} = g - \frac{k}{m} (e+x)$

c) As $ke = mg$

$$\ddot{x} = g - \frac{ke}{m} - \frac{k}{m}x = g - \frac{mg}{m} - \frac{k}{m}x$$

$$\Rightarrow \ddot{x} = -\frac{k}{m}x \quad \text{so} \quad \ddot{x} = -\omega^2 x$$

$$\text{for } \omega = \sqrt{\frac{k}{m}}$$

d) $k = 50$ ^{← stiffness}, $m = 2$, $x(0) = 1.5$, $\dot{x}(0) = 0$

$$\text{As } \omega = \sqrt{\frac{50}{2}} = \sqrt{25} = 5$$

so have

$$\rightarrow x(t) = A \cos(st) + B \sin(st)$$

could solve $\ddot{x} = -\omega^2 x$ to get auxiliary
 $m^2 + \omega^2 = 0$ $m = \pm \omega i$ so $x = A \cos \omega t + B \sin \omega t$.
^{recall solution to SHM}

$$\text{Now } x(0) = 1.5 \Rightarrow A = 1.5$$

$$\text{So } x(t) = 1.5 \cos 5t + B \sin 5t$$

$$\dot{x}(t) = -7.5 \sin 5t + 5B \cos 5t$$

$$\dot{x}(0) = 0 \Rightarrow 5B = 0 \quad \text{so } B = 0$$

$$\text{Hence } x(t) = 1.5 \cos 5t$$

Q6. If $\omega = 4$ in SHM

$$x(t) = A \cos 4t + B \sin 4t$$

$$x(0) = 0.5, \quad \dot{x}(0) = 0.4$$

$\uparrow$ Initial displacement
(in m) $\uparrow$ Initial velocity

so $A = 0.5$

$$\dot{x}(t) = -2 \sin 4t + 4B \cos 4t$$

so $4B = 0.4 \quad B = 0.1$

so $x(t) = 0.5 \cos 4t + 0.1 \sin 4t$

For max convert to $R \cos(4t + \alpha)$

then $R \cos \alpha = 0.5, \quad R \sin \alpha = -0.1$

$$R^2 = 0.5^2 + 0.1^2 = 0.26 \Rightarrow R = \sqrt{0.26}$$

$$\tan \alpha = \frac{-0.1}{0.5} \quad \alpha = -0.197$$

$$\sqrt{0.26} \cos(4t - 0.197)$$

so max is $\sqrt{0.26}$ at $t = \frac{0.197}{4} = 0.049..$

Q7. $\ddot{\Theta} = -\frac{g}{L} \Theta$ $L=2$ $\Theta(0)=0$
 $\dot{\Theta}(0)=0.4$

so $\omega = \sqrt{\frac{g}{L}} = \sqrt{\frac{g}{2}}$

Then $\Theta(t) = A \cos\left(\sqrt{\frac{g}{2}} t\right) + B \sin\left(\sqrt{\frac{g}{2}} t\right)$

$\Theta(0)=0 \Rightarrow A=0$

$\dot{\Theta}(t) = \sqrt{\frac{g}{2}} B \cos\left(\sqrt{\frac{g}{2}} t\right)$

$\dot{\Theta}(0)=0.4 \Rightarrow \sqrt{\frac{g}{2}} B = 0.4$

so $\dot{\Theta}(t) = 0.4 \cos\left(\sqrt{\frac{g}{2}} t\right)$

Q8. Motion does not continue for ever at the same amplitude, as energy is lost / damping occurs.

Model $\ddot{x} + \underline{bx} + \omega^2 x = 0$ with $b^2 - 4\omega^2 < 0$
 Linear damping and $b > 0$

Q9. λ is young's modulus of spring
elasticity

Q10.

Solution to $\ddot{x} = -\omega^2 x$ is

$$x(t) = A \cos \omega t + B \sin \omega t$$

with $x(0) = x_0 \Rightarrow A = x_0$

$$\dot{x}(t) = -\omega A \sin \omega t + \omega B \cos \omega t$$

with $\dot{x}(0) = v_0 \Rightarrow \omega B = v_0 \Rightarrow B = \frac{v_0}{\omega}$

So $x(t) = x_0 \cos \omega t + \frac{v_0}{\omega} \sin \omega t$

Now If $x(t) = R \sin(\omega t + \alpha)$

$$R \sin \alpha = x_0 \quad R \cos \alpha = \frac{v_0}{\omega}$$

$$R^2 = x_0^2 + \frac{v_0^2}{\omega^2} \quad R = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$$

$$\tan \alpha = \frac{x_0}{\frac{v_0}{\omega}} = \frac{\omega x_0}{v_0} \Rightarrow \alpha = \tan^{-1} \left(\frac{\omega x_0}{v_0} \right)$$

Amplitude: $\sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$, Period: $\frac{2\pi}{\omega}$, Phase: $\tan^{-1} \left(\frac{\omega x_0}{v_0} \right)$